

STUDY ON CLOUD COMPUTING AND CLOUD DATABASE

Manjeet Singh

School of Computing Science and Engineering
Galgotias University, G. Noida, U.P., India
er.manjeet106@gmail.com

Abstract— Cloud computing is basically to perform computation on a centralized facility provided by third party. It also provides storage facilities operated by the third party. Cloud computing allows us to hire entire platform for software development, hardware resources or setup a platform in house. Cloud computing offers everything as-a-service. It provides infrastructure as-a-service, platform as-a-service, software as-a-service and human as-a-service. This paper gives an introduction to what the cloud computing is, services it provide, how it is deployed, with its features and challenges. This paper focus on database management in cloud computing. It also analyses cloud DBMS characteristics, challenges and also compare various types of cloud databases.

Keywords— *Cloud Computing, Services, Cloud Database, Hadoop, MapReduce*

I. INTRODUCTION

The term *Cloud Computing* came into lime light in 2006, after Amazon's Elastic Computing Cloud (EC2) enters the world [1].

As per the National Institute of Standards and Technology (NIST) [2]: "Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction." Fig.1 shows framework of a cloud computing environment. NIST divides cloud computing into three "as-a-service" offerings, namely software infrastructure, and, platforms which are broken down in more detail in stack below. Cloud computing is providing "everything-as-a-service" (XaaS) [3].

A. Infrastructure as a Service (IaaS)

Cloud computing provide infrastructure as a service. Service provider offers control over different resources and appications to the organizations. Resources inclcludes storage devices, hardware, servers, networking components and any more.

B. Platform as a Service (PaaS)

This part of cloud model consist of a set of software and tools used for product development that allow developers to create applications on the provider's platform, i.e., applications can be built on the internet and executed on provider's infrastructure. This services is divided into two

parts, one is Programming Environments and another is Execution Environments.

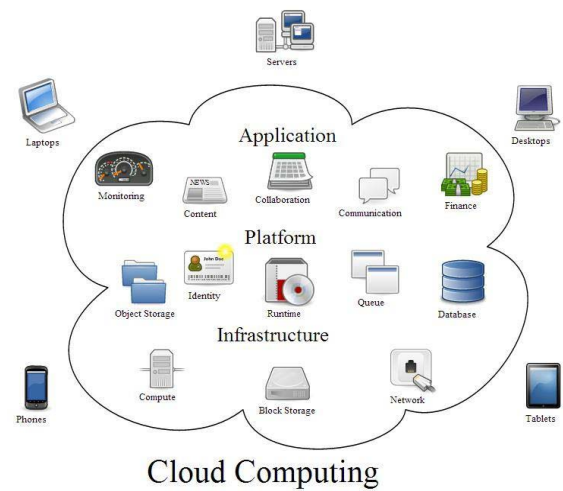


Fig.1. Cloud Computing Framework [26]

C. Software as a Service (SaaS)

This is the most familiar and prolific cloud service. All the applications that provide a direct service to the client and run on the Cloud are located in the SaaS layer.

II. CLOUD DEPLOYMENT MODELS

Depending upon the service provider there are various type of clouds, each having its own merits and demerits [4]. For example, one class of service providers is paying attention in reducing operation cost, while others may focus high reliability and security. Various possible models are Public Cloud, Private Cloud, Hybrid cloud and Community cloud.

A. Public cloud

The cloud in which resources are offered as services to the general public by service providers are known as public cloud. Several key benefits of using public clouds are no initial money investment on infrastructure and shifting of risks to infrastructure providers. But these clouds are deficient in fine-grained control over data, network and security settings, which hamper their usefulness in many business cases.

B. Private cloud:

Private clouds are designed for restricted use by a single association. It is also called internal cloud. It offers the uppermost degree of control over performance, reliability and security but may be criticized for being similar to traditional proprietary server farms and for not providing reimbursement such as no up-front capital costs.

C. Hybrid cloud

Hybrid cloud is a mixture of private and public cloud models that concentrate on the drawbacks of both models. In a hybrid cloud, one part of service infrastructure run in private cloud while another part runs in public cloud. Particularly, it provides tighter control and security over application data compared to public clouds, while still facilitating on-demand service expansion and contraction.

D. Community cloud

Community cloud allows sharing of computing infrastructure in between organizations of the same society. For example all administrative organization within Haryana, India may share computing infrastructure on the cloud to manage data related to general public residing in India.

III. CHALLENGES IN CLOUD COMPUTING

Cloud computing provide us several interesting features that makes it very pretty. For example, there is no up-front investment, also lowers operating cost, provide high scalability, easy access, reduced business risks and maintenance expenses etc [4].

The point to be noted is that while providing these features cloud computing has to face several challenges. Many existing issues related to it have not been completely resolved, while new challenges keep growing. This section summarizes some of the challenging research issues in cloud computing [4].

A. Automatic service provisioning

One of the major characteristics of cloud computing is its capability of acquiring and releasing resources on-demand. The purpose of a service provider is to assign and de-allocate resources from the cloud to satisfy its service agreement. However, it is not clear how a service provider can handle this challenge.

B. Virtual machine migration

By enabling virtual machine migration to balance load across the data centre we can be provided significant benefits in cloud computing via virtualization. By machine migration we can avoid hotspots. But detecting workload hotspots and initiating a migration lacks the ability to respond to sudden workload changes.

C. Server consolidation

It is used to maximize resource utilization while minimizing energy. Virtual machine technology is used to consolidate virtual machines running on various under-utilized servers onto a single server for energy-saving. The problem of

optimally consolidating servers in a data centre is NP-hard optimization problem [6].

D. Improving energy efficiency

It has been projected that 53% of total data center operational cost is used for powering and cooling [2]. A key challenge is to maintain a balance between application performance and energy savings.

E. Data security

Cloud service providers virtual machine technology does not have access to the physical safety system of data centers; they must rely on the infrastructure provider to attain full data security [7].

F. Storage technologies and data management

Cloud databases provide MapReduce like frameworks for example Hadoop and Dryad that are designed for distributed processing of data-intensive tasks. They lack various features of modern DBMSs like bulk loading, indexing, updating, integrity and referential constraints etc.

IV. DATABASE MANAGEMENT IN CLOUD COMPUTING

Database management applications are most likely elements for deployment in the cloud. Because enterprise database system comes with a large up-front cost, both in term of hardware and in software [8]. Cloud computing provides access to large pools of data and computational resources through a simple and combined interface between retailer and customer, allowing retailer to focus more on the software itself rather than the underlying framework.

A. Cloud Data Base Management System (CDBMS)

There are several ways in which computational capability and data storage facilities are provided to users. Users search for resources based on a variety of characteristics and provide access to large pool of data and computational resources through a variety of interfaces [9, 10]. Cloud Database Management system (CDBMS) is a distributed database that delivers computing as a service instead of product.

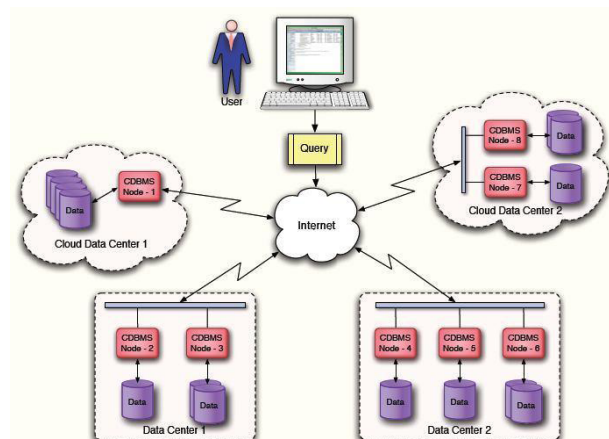


Fig.2. Cloud DBMS (CDBMS) [11]

The CDBMS pass the query traffic through different nodes. The Fig.2 [11] below illustrates that a query can be originated from anywhere through internet which is resolved by the cloud and corporate data centers via internet which are geographically distributed.

B. Challenges in CDBMS

Database is a system that provides a way to interact with data in an easy and managed manner. However, they have several serious limitations [11] associated with it.

1) *Scalability*: As the size of data grows, most database system does not easily scale beyond a certain limit. As with the passes of time the size of data is growing very fast and hence scalability is a major challenge [12].

2) *Deployment*: It is still a big challenge to create a database appliance that can be easily deployed in a cloud [13]. These issues may be related to localization, routing, authentication, placement of virtual machines on physical machines, how to handle varying workloads.

3) *Diversity in available systems*: The rise of specific database systems for particular markets complicates system selection, particularly for users whose requirement do not fall into one category.

4) *How to manage data in a secure way*: Data security is a major issue in cloud because data is managed by a third party and confidentiality of data becomes a major concern over here. The chances that host company may violate the privacy of its customers and access data without permission, makes some potential customers worried [7]. So, data security also becomes an important issue in cloud data management [14].

5) *Performance*: Performance is the first parameter that commercial database systems use to differentiate themselves from other solutions [15].

C. Characteristics of CDBMS

Following are the main characteristics for a cloud DBMS [7]:

1) *Efficiency*: Cloud DBMS should be efficient such that it can be used upto the maximum possible extent and the user have to pay for only what he use.

2) *Fault Tolerance*: Fault tolerance is the ability of a system to recover/restart from the nearest possible stable state after failure. Whenever a failure occurs, we have to recover from the failure (by choosing some mechanism like checkpointing etc.). Every time a node fails if we have to restart our query from the beginning then it becomes very hard for the long queries to complete. The only solution is fault tolerance, hence a system must be fault tolerant.

3) *Heterogeneous environment*: The environment of cloud computing is heterogeneous because the performance of various computing nodes is not consistent, some nodes deliver very high performance while others may not. If the work to be performed is equally divided amongst the cloud compute nodes, then there is possibility that some nodes may finish its task within times while others takes too much time. A node having degraded performance may thus have a

disproportionate affect on total task latency [17]. A database system that will run in such a heterogeneous environment must take appropriate dealings to stop this from happening.

4) *Data Encryption*: To provide authorized access only to the crucial data, a CDBMS must have an encryption scheme [18].

5) *Business intelligence support*: There is a number of tools for assistance in the query generation, visualization, and advanced data analysis etc.

D. Cloud DBMS types based on type of data

There are generally two types of cloud DBMS based upon the type of data stored in cloud [7]:

1) *Transactional data management*: The database applications that depends on ACID properties of transactions strongly and are fairly write-sensitive are known as transactional data management. These are the main source of income and existence of database industry. Applications like banking, reservation, e-commerce require transactional data management. These databases include Oracle, Microsoft SQL server, IBM DB2 and Sybase etc. These applications are not easily deployed in the cloud due to some issues like unavailability of shared-nothing architecture, it is difficult to maintain ACID guaranteed in because data is replicated in cloud over large geographical distances and security risks in storing data to an un-trusted host.

2) *Analytical data management*: When data store are queried for use in business planning, problem solving and decision support such type of data management is known as Analytical data management. Data from different databases is used in such type of analysis. Size of data in analytical data management is also considerably large as compared to transactional data management. Analytical DBMS are well-suited to cloud environment because of a number of reasons among which some are like they provide shared architecture, ACID guarantees are not needed typically and sensitive data is not kept for analysis.

On the basis of details given in [12, 13], the Table 1 compares the transactional and analytical Databases:

Table 1: Comparison of Transactional and Analytical Cloud Database

Properties	Transactional	Analytical
Source of Data	Operational data, it is original source of data	Consolidation data, data comes from transactional databases
Purpose of the data	To control and run primary Business job	To assist with plan, problem solving and judgment making
Processing speed	Usually very fast	It depends on the quantity of data involved
Space requirements	Comparatively small if historical data is logged.	Larger due to aggregation structures and history.
ACID guarantee	Hard to maintain,	Easy to maintain
Shared nothing architecture	Not a good match	Is a good match
Sensitive data	Risk to store transactional data	Can be left out of analysis

Above table shows that analytical systems are good candidate for deployment of data in the cloud because they do not create a great problem at the data level while relational databases require a centralized architecture which poses a scalability limit in data centers.

E. Cloud DBMSs as database-as-a-service providers

CDBMS as database-as-a-service makes various geographically distributed databases available and share them with various users and database logically function in such a way like all the databases are available local to the user. Database-as-a-service provider includes SimpleDB from Amazon, and Azure from Microsoft [21, 22].

F. Virtualization

Virtualization [23] is basically software technology which is drastically changing the information technology landscape and the way that people compute. Initially the computer hardware was designed to run a single operating system and a single application that leaves most of the machine underutilized. So for better utilization of resources a new technology comes into the focus, named as virtualization. It provides a dummy environment to perform the tasks in parallel with the best possible use of available resources.

1) *Virtualization and Cloud DBMS* : The Cloud provides the services in the distributed environment which requires the availability of the multiple machines. The virtualization enables us to work on multiple virtual machines on a single physical system. So, virtualization helps the researchers to analyze the cloud environment on single machine which leads to reduction in cost, increase the availability, performance and better analysis.

The discussion of cloud DBMS shows that managing the data on the cloud environment has several challenges as the database systems are difficult to scale, configure, maintain and increases the cost for provisioning of peak time load. The transactional database applications are not a severe problem but the analytical and relational database would encounter the scalability limit at some point. The Parallel DBMS lacks in Fault tolerance, ability to work in heterogeneous environment and support for Business Intelligence and data encryption while the MapReduce are not as much efficient and does not provide data encryption support as provided by the Parallel DBMS. So, there is a need for Hybrid architecture which can combine the properties of both the tools available. The most of the Database-as-a-Service providers are not Open Source, while Apache Hadoop is open source project which can be used for research purpose to analysis of database applications in the cloud environment. Virtualization helps to create a cluster of nodes on a single machine which enhances the performance of the application by fully understanding the best practices on virtualized infrastructure.

G. Apache Hadoop

Apache Software Foundation develops a java based open source s/w platform called Hadoop. Hadoop implements Google's MapReduce programming model on top of a

distributed file system called the Hadoop Distributed File System (HDFS)[22].

Hadoop is an s/w platform which helps in easily writing applications for processing vast amounts of data in-parallel on large clusters of commodity hardware in a reliable, fault-tolerant manner. A Map/Reduce job generally divides the input data-set into independent parts which are processed by the map tasks parallel. Hadoop sorts the outputs of the maps, which are then input to the reduce tasks. Both I/O of the job are stored in a file-system. The framework takes care of scheduling tasks, monitoring them and re-executes the failed tasks. The Map-Reduce framework consists of a single master Job Tracker and one Slave Task Tracker per cluster-node. The master takes care of scheduling, monitoring and re-executing the tasks.

H. Cloud DBMS Tools

Cloud DBMS tools can be classified into two categories one is parallel DBMSs and other is MapReduce based software's:

1) *Parallel DBMS*: These databases started to develop in 1980s [17]. Parallel DBMS support standard relational tables and SQL and also implements many of the performance enhancing techniques including indexing, compression, materialized views, result caching and I/O sharing and so on.

2) *MapReduce based Softwares*: MapReduce was introduced in 2004 by Dean et al. It is a s/w model for processing and generating large data-sets that is amenable to a large variety of real-world tasks. Clint gives input in form of map and a reduce function and the underlying runtime system automatically parallelizes the computation [16].

Table 2: Comparison of Parallel DBMS and MapReduce Framework

Properties	Parallel DBMS	MapReduce
Efficiency	Complex when the grows	Dependent upon application
Heterogeneity	Designed to run on homogeneous environment	Designed to run on heterogeneous environment
Fault Tolerance	Need to restart a query upon failure	Taken as high priority
Business Intelligence Interface	Does not easily interface	Carefully optimized and certified BI interfaces
Data Encryption Support	Works with hand coded encryption	Unable to operate on encrypted data

A cloud DBMS should meet the characteristics as described above. Based on these cloud characteristics parallel DBMS and MapReduce framework are compared as shown in Table 2 [7, 10, and 12].

V. DATA ENCRYPTION IN CLOUD DBMS

In cloud computing framework data security becomes an important issue due to so many cloud applications being available. I consider data confidentiality in CDBMS. I assume these storage servers are sincere for storage services but malevolent for data confidentiality. Data confidentiality become major concern when storage servers are owned by a

cloud infrastructure provider (public cloud) and data are owned by other parties [25]. The problem of data security in Cloud computing become bottleneck of cloud computing [1]. Due to the shared and distributed environment of the cloud, the need for ensuring security of its critical infrastructure that includes computing, network and storage is very important [14]. One of the significant resources to look after in cloud environment is the data which is stored in files. The files can be of any type, it may be configuration file at servers, or private user confidential file at users own work place but they all have a danger of data modification linked with them. Security is one of the key issues which obstruct the growth of cloud. The idea of handing over important data to another company is bothersome; such that the consumers need to be vigilant in understanding the risks of data breaches in these new surroundings [25]. In cloud computing security there are mainly three types of issues:

A. Service Provider Issues

Cloud service provider should ensure organizational security and privacy needs.

- How to manage permission, verification and auditing of users accessing cloud services.
- How to manage the privacy of confidential data while performing different operations on such private data.

Data can be changed while is in transition. So it is also the responsibility of cloud service provider where the data should go or not using proper encryption schemes.

B. Infrastructure Security Issues

Cloud supplier provides varying services to different type of clients so a security aspect also varies for different customers. Infrastructure Security Solutions are easy to implement, however infrastructure also have some issues [25]:

- How to ensure secure data storage which concerns the way in which data is accessed and stored, auditing of data, segregation of sensitive data.
- Virtual servers and applications needs to be secured each physically and logically.
- Preventing holes or leaks between the composed infrastructures is a major concern with hybrid clouds.

C. End-user-security Issues

End Users need to access resources within the cloud and may bear in mind of access agreements like acceptable use or conflict of interest. Some issues related to their security are:

- Providing security-as-a-service by using either by using established information security vendors or by cloud service providers providing security as cloud service with information security companies.
- How to browser security for different Input/output operations, for authorization and authentication operations.
- There is a loss of governance while client uses cloud services so leaving gap in security defenses.
- There is a problem of data lock-in when providing service portability by standard formats or service interfaces.

How to check data protection when handling data to cloud service providers by clients?

VI. CONCLUSION

In this paper I study the cloud computing and cloud database. Paper explains various services provided by cloud computing, its deployment models, features and challenges. Then I analyze cloud database is in details with various challenges it faces like scalability deployment, diversity, security and performance. In conclusion I can say that cloud computing is a centralized facility operated by a third party to perform computation and to provide storage utility. Cloud computing provides everything as a service. Cloud computing offers access to data base and various resources needed for computation through a simple and unified interface between vendor and user. A CDBMS should be efficient, fault tolerant, secure and should support business intelligence.

I compare transactional DBMS and analytical cloud database over various properties like source of data, processing speed, ACID guarantee etc. and analyze strength and weakness each of them. Cloud DBMS also acts as-a-service provider, for example: Amazon simpleDB and Microsoft Azure. This paper also uncover the Apache Hadoop database and the cloud DBMS with virtualization. On comparing parallel DBMS and MapReduce framework cloud databases I come to conclude that: (1) in parallel DBMS maintaining efficiency is complex when database grows; (2) Parallel database are suitable only for homogeneous system; (3) fault tolerance maintenance is hard (need to restart the query upon failure); (4) MapReduce framework are unable to operate on encrypted data.

VII. FUTURE WORK

In future I will be developing a secure and efficient approach for database management in cloud computing by combining the better attributes of parallel DBMS and MapReduce framework cloud database.

REFERENCES

- [1] Yuefa D., Wu Bo, Yaqiang G., Quan Z., and Chaojing T., "Data Security Model for Cloud Computing," in *Proce.of the 2009 IWSA*, China, November 21-22, 2009.
- [2] P. Mell and T. Grance, "The NIST definition of Cloud Computing," National Institute of Standards and Technology, Information Technology Laboratory, Version 15, July 2009.
- [3] Lenk A., Klems A., Nimis J., Tai S., and Sandholm T., "What's inside the cloud? An architectural map of the cloud landscape," in *CLOUD '09: Proceedings of the 2009 ICSE Workshop on Software Engineering Challenges of Cloud Computing*, Washington, DC, USA. IEEE Computer Society, 2009.
- [4] Qi Z., Lu C., and Raouf B., "Cloud computing: state-of-the-art and research challenges", Springer, pp 7-18, Apr 2010.
- [5] Armbrust M et al, "Above the Clouds: A Berkeley View of Cloud," in *EECS*, Technical Report No. UCB/EECS-2009-28, University of California, Berkeley, February 10, 2009.
- [6] Branter M., Graf D., and Florescu D. et al, "Building Database in the cloud", in *Systems Group*, ETH, ZURICH.
- [7] Daniel J., Abadi, "Data Management in the Cloud: Limitations and Opportunities," Yale University, New Haven, CT, USA. 2009.
- [8] Buyya R., "Market-Oriented Cloud Computing: Vision, Hype and Reality for Delivering IT Services as Computing Utilities," 2008.

- [9] Tuan V., "Cloud Data Management," IFSIC, IRISA, Ker Data Project-Team. Jan, 2010.
- [10] Chris G. et al., "The Eucalyptus open source cloud computing system," IEEE International Symposium on Cluster Computing and the Grid, 2009.
- [11] Robin B., "What is Cloud Database? Suitability of Algebraic Data's Technology to Cloud Computing", The Bloor Group, White Paper, 2011.
- [12] Das S., Elmore A., Wang S., Agrawal D., Abbadi A. E., "Autonomic, Elastic, Fault-tolerant, Scalable, and Secure Data Management in the Cloud," University of California, CA 93106-5110, Santa Barbara, USA.
- [13] Ashraf A. et al., "Deploying Database Appliances in the Cloud," Data Engineering, IEEE Computing Society, 2009.
- [14] Sanchika G. et al., "A Secure and lightweight approach for Critical Data security in Cloud," in Fourth International Conference on Computational Aspects of Social Networks, 2012.
- [15] Abouzeid A. et al., "HadoopDB: An Architectural Hybrid of MapReduce and DBMS Technologies for Analytical Workloads," in VLDB, 2009.
- [16] Dean J. and Ghemawat S., "MapReduce: Simplified data processing on large clusters," in Proc. of 6th USENIX Symposium on Operating Systems Design and Implementation, OSDI 2004, San Francisco, USA, Dec. 2004.
- [17] Buyya R. et al., "Cloud Computing and Emerging IT Platforms: Vision, Hype, and Reality for Delivering Computing as the 5th Utility," Future Generation Computer Systems", Elsevier Science, Amsterdam, Netherlands, June 2009.
- [18] Kantarcoglu M. and Clifton C., "Security issues in querying encrypted data," in 19th Annual IFIP WG 11.3 Working Conference on Data and Applications Security, 2004.
- [19] Pavlo A. et al., "A Comparison of Approaches to Large scale Data Analysis," in Proc. of SIGMOD, 2009.
- [20] Liladhar R., Rewatkar and Ujwal A., "Data management in market oriented cloud," Advances in Computational Sciences and Technology, ISSN 0973-6107 Volume-3, 2009.
- [21] <http://aws.amazon.com/simplifiedb/>
- [22] <http://www.microsoft.com/windowsazure/>
- [23] www.aasysgroup.com "Virtualization Basics," AaSys Solutions, Issue 9, Volume 6, September 2008.
- [24] Gurudatt K. et al., "A Security Aspects in Cloud Computing," in IEEE 3rd international conference on S/W Engineering and Service Science (ICSESS), pp. 547-550, ISBN 978-1-4673-2007-8, Beijing, June 2012.
- [25] Krishna V. R. et al., "Security Architecture of Cloud Computing," International Journal of Engineering Science and Technology (IJEST), Vol. 3 No. 9 September 2011.
- [26] <http://www.techbyter.com/2013/0224.php#sthash.56qU91Qg.dpbs>